Introduction to Artificial Intelligence (AI)

ML libraries

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Machine learning libraries

1. Scikit-learn

•**Description**: A robust library for classical machine learning algorithms, providing simple and efficient tools for data mining and data analysis.

•**Key Features**:

- Supports classification, regression, clustering, and dimensionality reduction.
- User-friendly API and extensive documentation.
- Works well with NumPy and pandas.

Machine learning libraries

2. TensorFlow

•**Description**: An open-source library developed by Google for deep learning and neural network research.

•**Key Features**:

- Supports a wide range of neural network architectures.
- High scalability and can run on multiple CPUs and GPUs.
- Offers TensorFlow Lite for mobile and embedded devices.

Machine learning libraries

3. Keras

•**Description**: A high-level neural networks API that runs on top of TensorFlow, simplifying the creation of deep learning models.

•**Key Features**:

- User-friendly and modular, making it easy to experiment with different neural network architectures.
- Supports convolutional and recurrent networks as well as combinations of both.

Machine learning libraries

4. PyTorch

•**Description**: An open-source machine learning library developed by Facebook, widely used for deep learning applications.

•**Key Features**:

- Dynamic computation graph, allowing for more flexibility during model training.
- Strong support for GPU acceleration and distributed training.

Machine learning libraries

5. Pandas

•**Description**: Although not a machine learning library per se, it's an essential data manipulation library for preparing data for machine learning.

•**Key Features**:

- Provides data structures for efficiently handling and analyzing data.
- Integrates well with other machine learning libraries.

Machine learning libraries

6. NumPy

•**Description**: A fundamental library for numerical computing in Python.

•**Key Features**:

- Provides support for arrays, matrices, and a host of mathematical functions.
- Often used in conjunction with other libraries.

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7. OpenCV

•**Description**: A library for computer vision tasks.

•**Key Features**:

- Supports image processing, video analysis, and real-time computer vision.

Machine learning libraries

8. Matplotlib

•**Description**: Libraries for data visualization.

•**Key Features**:

- Matplotlib: A plotting library for creating static, animated, and interactive visualizations.